

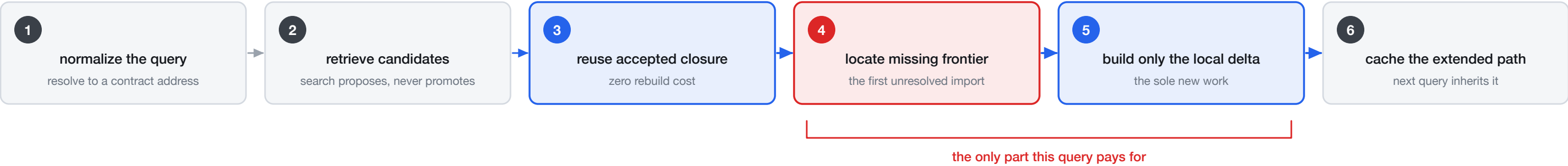
The registry behaves like a package manager with an incremental build

The expensive offline path standardizes contracts once. The online path reuses the accepted closure and pays only for the frontier that is actually missing.

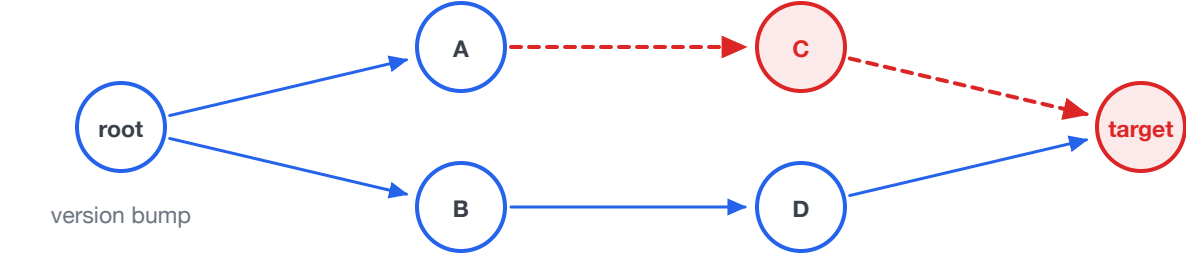
OFFLINE — done once, amortised over every later query



ONLINE — what a single query actually costs



WHEN AN UPSTREAM CONTRACT CHANGES



● invalidated — must be rechecked

● untouched — stays accepted, no rebuild

Invalidation follows the dependency edges, not the calendar.